DOST Form 2 (for Basic/Applied Research)

DETAILED RESEARCH & DEVELOPMENT PROJECT PROPOSAL

(1) PROJECT PROFILE

Program Title: N/A

Project Title: **Promoting Coastal Resource Conservation Activities by Local Communities through Fostering Social Capital: Approach Using Public Good Experiment**

Project Leader/Sex: Joela Mizchelle D. Dacillo, PhD/ Female

Project Duration (number of months): 24 months

Implementing Agency: Partido State University - College of Fisheries and Marine Science

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(2) COOPERATING AGENCY/IES (Name/s and Address/es)

- Department of Agriculture – Bureau of Fisheries and Aquatic Resources (DA-BFAR) II, Tuguegarao City, Cagayan, Philippines
- Kochi University, Kochi, Japan

(3) SITE(S) OF IMPLEMENTATION

IMPLEMENTATION SITES NO.	COUNTRY	REGION	PROVINCE	DISTRICT	MUNICIPALITY	BARANGAY
1.	Philippines	Region V	Camarines Sur	Fourth District	Sagñay	Atulayan
2.	Philippines	Region V	Camarines Sur	Fourth District	Sagñay	Turague
3.	Philippines	Region V	Albay	First District	Tiwi	Misibis
4.	Philippines	Region II	Cagayan	Second District	Santa Praxedes	Macatel
5.	Philippines	Region II	Cagayan	Second District	Santa Praxedes	San Juan
6.	Philippines	Region II	Cagayan	First District	Santa Ana	San Vicente

(4) TYPE OF RESEARCH

_____ Basic

__X__ Applied

(5) R&D PRIORITY AREA & PROGRAM (based on HNRDA 2017-2022)

__X__ Agriculture, Aquatic and Natural Resources
Commodity: Agricultural and resource economics assessment__

_____ Health
Priority Topic: _____

_____ Industry, Energy and Emerging Technology
Sector: _____

_____ Disaster Risk Reduction and Climate Change Adaptation

_____ Basic Research
Sector: _____

Sustainable Development Goal (SDG) Addressed

__Life Below Water_____

(6) EXECUTIVE SUMMARY (not to exceed 200 words)

The project aims to address the pressing challenges of sustainable coastal resource management in the Philippines, particularly in coastal areas near Marine Protected Areas (MPAs). These regions face growing threats from overexploitation and inadequate management, primarily due to limited community participation. To explore how social capital and cultural values influence cooperative behavior among indigenous and local fishers, the project will employ Public Goods Experiments (PGE). This applied research will be conducted in selected coastal communities within the Bicol Region and Cagayan Valley, comparing the behavior of indigenous and non-indigenous fishers to identify factors that enhance cooperation. The key objectives of the study include investigating the role of social capital in fostering cooperation, assessing the impact of cultural values on resource management, and analyzing socio-economic factors that affect collective action. Findings will inform policy recommendations to strengthen community engagement and promote sustainable marine resource management, contributing to the long-term resilience of both ecosystems and local communities. The project aligns with national sustainable development and environmental conservation priorities, aiming to provide actionable insights and evidence-based strategies for more effective MPA management and socio-economic resilience.

(7) INTRODUCTION

Marine resources are essential to the livelihoods of coastal communities, especially for local and indigenous fishers who depend on these resources for both sustenance and economic well-being. However, the sustainability of these shared resources is increasingly threatened by overexploitation and environmental degradation, exemplifying the tragedy of the commons, where individuals acting in their self-interest, overexploit communal resources, ultimately leading to their depletion. Addressing these challenges requires effective marine resource management built on collective action, trust, reciprocity, and cooperation among stakeholders. This research aims to investigate the dynamics of collective action through a Public Goods Experiment (PGE) to promote sustainable marine resource management among indigenous and local fishers.

PGEs serve as valuable tools for understanding collective decision-making and cooperative behavior in the management of shared resources. By simulating scenarios where individuals decide how much to contribute to a communal fund, PGEs provide valuable insights into the factors that influence cooperation, such as trust, social norms, and group dynamics. In this study, the PGE is designed to explore how indigenous and local fishers cooperate under varying conditions, such as homogeneous and heterogeneous group compositions, and how repeated interactions influence contributions. This controlled yet realistic approach enables a deeper understanding of the role of social capital, cultural values, and group dynamics in marine resource management. Coastal resource conservation within and near MPAs is treated as a public good in this study due to its non-excludable and non-rivalrous benefits. This framework allows for an in-depth analysis of collective action, cooperation, and trust-building among local and indigenous fishers, which are crucial for sustainable resource management.

Social capital, encompassing elements such as trust, reciprocity, and social networks, plays a critical role in fostering cooperation among stakeholders in resource management. Trust facilitates information sharing and resource pooling, reciprocity strengthens community bonds through mutual support, and social networks provide platforms for collaboration and norm-setting. In shared marine environments, the presence of strong social capital encourages cooperation and collective decision-making, which are essential sustainable practices. This study will investigate the role of social capital among indigenous and local fishers, analyzing how its varying levels influence their willingness to cooperate and engage in conservation efforts. Through this lens, the research seeks to identify pathways to enhance collaboration and address barriers to sustainable management.

Cultural values also influence how communities perceive and engage in resource management. Indigenous communities often possess cultural practices and beliefs deeply rooted in sustainable interactions with their environment. These traditions can complement modern conservation strategies, fostering a shared understanding of resource stewardship. For instance, customary rituals, taboos, or collective fishing practices often embody principles of sustainability and respect for marine ecosystems. This study seeks to explore how the cultural values of indigenous and local fishers influence cooperative behavior, providing insights into how cultural factors can be leveraged to enhance sustainable marine resource management.

The focus on areas near the Kuroshio Current adds depth to this study. As one of the world's strongest ocean currents, it supports rich marine ecosystems and provides vital resources to coastal communities. However, these areas are highly vulnerable to overfishing and environmental changes, making sustainable management crucial. The strong cultural and economic ties of local communities to the ocean offer a unique setting to explore how social capital, cultural values, and cooperation influence resource management. Insights from these areas can guide conservation efforts in similar ecologically and culturally significant regions.

Focusing on fishers near marine protected areas (MPAs) influenced by Kuroshio Current, this study seeks to uncover how social capital and cultural values influence cooperative behavior in the context of marine resource management. By analyzing the effects of group composition, repeated interactions, and rewards through PGEs, this research aims to generate actionable insights for community-based management strategies, and policy recommendations that support sustainable practices.

Our previous surveys of the communities managing MPAs have highlighted the importance of social capital, particularly trust and reciprocity, in promoting participation in MPA management. These studies revealed that higher levels of trust and reciprocity correlate with greater willingness to work (WTW) for MPA management. However, questions remain about how internal community structures, cooperative labor, and shared norms further shape this willingness. At present, marine ecosystems face growing threats from unsustainable fishing practices. Coral reefs, fish stocks, and marine biodiversity are in decline, impacting livelihoods and food security. Coastal communities, especially local and indigenous fishers, also rely heavily on marine resources for survival and income. Without proper management, these resources may not be available for future generations. Moreover, MPAs aim to conserve biodiversity and support sustainable fishing. However, many MPAs struggle to achieve their goals due to weak community

engagement and cooperation. By addressing these gaps, the study aims to deliver targeted policy recommendations to foster social capital, enhance MPA management, and promote sustainable resource use.

By investigating the connections between social capital, cultural values, and cooperative behavior, this research will offer a deeper understanding of the social and cultural drivers of collective action in marine resource management. Using PGE, it will provide a robust framework to analyze trust, reciprocity, and traditional practices, aiming to inform policies and strategies that strengthen collaboration, promote sustainability, and empower coastal communities to protect their shared resources. Through the PGE, the team will craft specific policy strategies for the effective management of the MPA based on the individual and community behavior and its implications. The outcomes of the PGE will be used as inputs for the planning for the MPAs. The team will be engaging with the LGUs to disseminate results and integrate local inputs to the MPA management plans.

(7.1) RATIONALE/SIGNIFICANCE (not to exceed 300 words)

Marine resources are important to the livelihoods of coastal communities in the Philippines, yet they face increasing threats from overexploitation, environmental degradation, and ineffective management. MPAs have been established to address these challenges, but their success varies widely across regions. Key issues include insufficient community participation, the free-rider problem in shared resource management, and a lack of integration of local cultural values into conservation efforts. Without addressing these underlying problems – particularly the need for stronger social capital and cooperative behavior among local and indigenous fishers – MPAs will continue to underperform, jeopardizing both ecological sustainability and the economic well-being of these communities.

This proposal aligns with the Philippines' national priorities in Science and Technology (S&T), particularly in the areas of sustainable development, environmental conservation, and community empowerment. By focusing on the enhancement of social capital and cooperative management of MPAs, the project directly supports the country's strategic goals for sustainable marine resource management. The research is also strategically relevant to national development, as it aims to strengthen the resilience of coastal communities, contribute to food security, and promote inclusive growth. Furthermore, the project is sensitive to the Philippine context, respecting local traditions, cultural values, and gender dynamics. It seeks to empower both men and women in these communities, ensuring that conservation efforts are inclusive and culturally appropriate. This approach not only addresses immediate environmental challenges but also fosters long-term socio-economic stability in the Philippines.

This study directly addresses real-world challenges faced by coastal communities in the Philippines by integrating actionable strategies that emphasize community involvement, practical outcomes, and capacity building. It engages local and indigenous fishermen to ensure their cultural values and knowledge shape solutions. Partnerships with local governments and fisherfolk associations enable collaborative, community-driven MPA management. The research provides evidence-based recommendations for policy reforms, ensuring practical impact. Training workshops enhance social capital and cooperative skills, empowering stakeholders. Gender inclusivity ensures equitable participation, while monitoring mechanisms and practical tools ensure adaptability, scalability, and long-term ecological and socio-economic benefits. By anchoring the research in community needs, aligning it with policy recommendations, and focusing on actionable outcomes, this initiative goes beyond academic boundaries to foster resilient and sustainable coastal communities in the country.

(7.2) SCIENTIFIC BASIS/THEORETICAL FRAMEWORK

The scientific basis of this research is firmly anchored in the principles of PGEs and the inherent challenges in managing public goods, such as MPAs. Public goods are characterized by their non-excludability and non-rivalrous nature, meaning that individuals cannot be effectively excluded from use, and one person's use does not diminish their availability to others (Samuelson, 1954). These attributes often lead to the "free-rider problem", where individuals benefit from the resource without contributing to its maintenance, risking its depletion or degradation (Olson, 1965).

Empirical studies in experimental economics emphasize the importance of cooperative behavior for sustainably managing public goods. PGEs simulate real-world scenarios where individuals must choose between contributing to a communal resource or prioritizing self-interest. These experiments have consistently shown that factors such as social capital, communication, group size, and cultural values significantly impact cooperation levels within communities (Ostrom, 1990; Fehr & Gächter, 2000). For instance, higher levels of social capital – marked by trust, reciprocity, and robust community networks – have been shown to enhance cooperation in shared resource management (Putnam, 1993). Additionally, cultural values, particularly those embedded in indigenous communities, often promote sustainable practices and collective responsibility, which are critical for effective MPA management (Berkes, 2004).

The theoretical framework of this study integrates concepts from game theory, social identity theory, collective action theory, and behavioral economics to analyze and interpret data collected through PGEs and surveys.

- Game Theory: PGEs draw heavily on game theory, particularly the Prisoner's Dilemma, where participants decide between cooperation and defection. While defection may maximize individual gain, mutual cooperation results in superior collective outcomes (Axelrod, 1984). This theory underpins the analysis of participants' decision-making processes in the experiments.
- Social Identity Theory: This theory posits that individuals' behavior in cooperative settings is shaped by their group identification. A strong group identity fosters cooperation, as individuals prioritize the group's welfare (Tajfel & Turner, 1979). In this study, social identity theory explains the cooperative dynamics among different ethnic and mixed groups within the communities.
- Collective Action Theory: Elinor Ostrom's framework explores how communities overcome collective action challenges in managing common-pool resources. It highlights the role of local rule-making, effective monitoring, and graduated sanctions in sustaining cooperation (Ostrom, 1990). This theory is vital for understanding the interplay between social capital, governance structures, and MPA management.
- Behavioral Economics: This perspective recognizes that economic decisions are influenced by factors beyond rational self-interest, such as fairness, reciprocity, and social norms (Thaler, 1980). Behavioral economics sheds light on how these elements shape cooperative behavior, particularly in culturally diverse settings.

This integrated approach provides a robust foundation for exploring the dynamics of social capital, cooperative behavior, and sustainable resource management in MPAs.

(7.3) OBJECTIVES

General:

This study generally aims to examine the factors influencing cooperative behavior among indigenous and local fishers in the Philippines for sustainable coastal resource management.

Specific:

1. To measure and compare the cooperative behavior among indigenous and local fishers through the public goods experiment in the context of sustainable coastal resource management.

2. To analyze the effect of social capital (levels of trust and reciprocity) on cooperative behavior among indigenous and local fishers in coastal resource management.

3. To assess how cultural values (beliefs about communal responsibility, respect for nature, attitude towards marine resource sharing and management, etc.) of indigenous and local fishers influence their social capital and cooperative behavior in coastal resource management.

4. To examine the influence of socio-economic factors (age, income, educational level, fishing experience, etc.) on social capital and cooperative behavior in coastal resource management.

5. To provide recommendations for promoting sustainable coastal resource management, especially within MPAs, by enhancing social capital and fostering cooperative behavior, considering the role of incentives and the influence of cultural values.

(8) REVIEW OF LITERATURE

The promotion of coastal resource conservation through fostering social capital has gained increasing attention as researchers have recognized the importance of community-based approaches in sustainable environmental management. Social capital, which encompasses the networks, norms, and trust that facilitate coordination and cooperation for mutual benefit (Putnam, 1995), is integral in encouraging community participation in conservation efforts. Studies have shown that communities with strong social capital are more inclined to participate in collective action, achieving more effective and sustainable resource management outcomes (Pretty & Ward, 2001). In the context of coastal resource conservation, leveraging social capital enhances community resilience, promotes shared responsibility, and fosters long-term stewardship of natural resources (Gutiérrez et al., 2011). The proposed study uses a Public Goods Experiment (PGE) to investigate how social capital can be cultivated to support conservation initiatives in coastal communities.

Ecological and environmental factors such as those influenced by the Kuroshio Current, a major western boundary current, significantly influence coastal resource systems and community behaviors. The current's nutrient-rich waters sustain biodiversity and fisheries productivity in coastal areas, creating hotspots of marine activity (Ando et al., 2021). Seasonal variations in the current influence the distribution and abundance of fish stocks, requiring fishers to adapt through collective strategies (Saito, 2019). Such

ecological dynamics underscores the necessity of the formation of robust community cooperation for effective shared resources management.

Indigenous knowledge systems, deeply rooted in centuries of ecological observations and sustainable practices, complement modern conservation methods. These systems, often embodied in unwritten "rules of the sea," ensure equitable resource access and minimize conflicts, reflecting cultural values passed down through generations (Berkes, 1999). Such traditions promote collective responsibility and stewardship, aligning with conservation objectives. For instance, Alcala and Russ (2006) highlighted the success of community-managed MPAs in Apo Island, where traditional practices contributed to the recovery of fish stocks and coral reefs. Similarly, the Kuroshio Current fosters cooperative behaviors and knowledge sharing in resource management, alongside governance mechanisms that resolve resource-related conflicts (Liu et al., 2021; Kanaji et al., 2022).

The influence of modernization often disrupts traditional practices, leading to the loss of valuable ecological knowledge (Menzies & Butler, 2007). Younger generations may prioritize economic gains over traditional practices, necessitating educational initiatives to preserve TEK (McCay, 2002). Alcala and Russ (2006) demonstrated the success of community-managed MPAs in Apo Island, where collective efforts to regulate fishing have led to the recovery of fish stocks and coral reef health. The involvement of local traditions and knowledge was key to the program's success. Similarly, the cultural and behavioral dynamics shaped by the Kuroshio Current promote the sharing of knowledge and collaborative approaches to resource management (Liu et al., 2021). Additionally, local governance mechanisms influenced by this current play a crucial role in resolving resource-related conflicts (Kanaji et al., 2022).

Ito et al. (2017) used a Philippine MPA database and a management effectiveness evaluation tool to evaluate a large number of MPAs and concluded that less than half of the MPAs are effectively managed. Based on this, they pointed out the need to conduct more detailed field research on individual MPAs, but most of these studies are natural science studies, and there has been insufficient research on social science analysis of the underlying factors in the local community. It is important to empirically analyze the social structure and management organization of the community, and the motivation of local residents to participate, and to study how to achieve effective management in order to promote the conservation of coastal natural resources. For the long-term sustainable conservation and management of MPAs, it is important to form social capital in the community involved in the management of the MPA, along with an institutional framework to back it up. For example, it is necessary to clarify the social capital embodied in the local social network that contributes to promoting the activities of the organization, as proposed by Schneider (2009) using the concept of organizational social capital. To do this, it is necessary to clarify the relationship between the social capital and the community and normative consciousness of the residents that underpins it, based on the internal structure of the community.

Public Goods Experiments (PGEs) are widely used in experimental economics to study cooperation and collective action in managing common-pool resources. In the context of coastal resource management, these experiments provide valuable insights into the behavioral dynamics of communities reliant on shared marine resources. PGEs are rooted in the economics of public goods, characterized by non-excludability and non-rivalry (Samuelson, 1954). In the context of marine resources, these experiments simulate decision-making scenarios where individuals decide whether to cooperate (contribute to the public good) or free-ride (benefit without contributing). The free-rider problem (Olson, 1965) is a key challenge in managing public goods, particularly in shared resource systems like fisheries. PGEs often employ game-theoretic models such as the Prisoner's Dilemma to analyze cooperation. Fehr and Gächter (2000) highlighted the role of social preferences, fairness, and reciprocity in influencing contributions to public goods, providing a behavioral foundation for designing resource management strategies.

(9) METHODOLOGY

This research will employ a Public Goods Experiment (PGE) to explore the dynamics of social capital and cooperative behavior among indigenous and local fishers in selected regions of the Philippines. This experiment will simulate the collective management of a shared marine resource in the study areas. The study will be structured to assess the impact of cultural values, social capital, and rewards on decisions of participants to contribute to a communal resource, with a particular focus on the comparison between different group compositions and the introduction of rewards.

Study Sites and Context: The study will focus on Sagñay (Atulayan and Turague), Tiwi (Misibis), Sta. Praxedes (Macatel and San Juan), and Santa Ana (San Vicente), locations selected for their ecological richness, cultural diversity, and logistical accessibility. Sagñay and Tiwi are municipalities in the Bicol Region, while Sta. Praxedes and Santa Ana are municipalities in Cagayan Valley. These areas, situated adjacent to MPAs and influenced by the Kuroshio Current, provide high-biodiversity contexts and diverse fisher groups for analyzing cooperative dynamics and cultural interplay in coastal resource management. The varying levels of social capital within these communities make them ideal for investigating trust and reciprocity in promoting coastal resource conservation. Established collaborations and accessible

infrastructure ensures the feasibility of fieldwork while reinforcing their relevance to examining the interaction between social capital, cultural values, and cooperative behavior in coastal resource conservation.

Partnership with DA-BFAR: Key support will come from DA-BFAR regional office, which will play a critical role in essential coordination, technical expertise, and capacity-building initiatives. As a partner agency, DA-BFAR will facilitate connections between the research team, local government units, and fishing communities to ensure seamless field implementation. Their expertise in fisheries management and MPA operations will guide the study, while their logistical support will include organizing community meetings, workshops, and data collection activities. Acting as a bridge to policymakers, DA-BFAR will ensure that the findings of the study align with regional and national fisheries strategies. Their involvement is indispensable, as their mandate and established relationships with local stakeholders ensure trust, smooth implementation, and impactful integration of research outcomes into broader conservation efforts.

Conceptual Framework: Figure 1 illustrates the study's conceptual and methodological design for understanding cooperative behavior of local and indigenous fishermen in coastal resource management. It displays the hypothesized relationships among core variables: the social capital (SC) which refers to trust, reciprocity and networks among individuals or groups that could influence cooperative behavior; the cultural values (CV) which represents the shared beliefs, tradition and practices that affect attitude toward cooperation and coastal resource management; the socio-economic factors (SE) which include economic status, income levels, education, and other demographic factors that might directly or indirectly influence cooperation; and the cooperative behavior (CB) which represents the actions taken by individuals or groups that are intended to achieve mutual benefit or a common goal, often requiring collaboration, sharing, or joint effort.

The study will employ a three-stage data-collection process: (1) Before PGE: baseline surveys to gather data on SC, CV and SE attributes of the participants; (2) During PGE: behavioral data collection based on participants' contribution levels to communal fund, reflecting CB in a simulated setting; and (3) After PGE: post-experiment surveys and focus group discussions to assess changes in perceptions, attitudes, and SC influenced by the PGE.

In this framework, the outcomes and benefits of coastal resource conservation efforts are classified as public goods because they are non-excludable and non-rivalrous. Once conservation efforts are implemented, their benefits (e.g., cleaner water, restored fish stocks, protection from storm surges, and biodiversity preservation) are available to everyone in the community, regardless of whether they contributed to the efforts. These benefits cannot be restricted to specific individuals, making them non-excludable. Additionally, one person's use or enjoyment of these benefits does not reduce their availability to others, making them non-rivalrous. This public good nature underscores the importance of collective action and cooperation to sustain these shared benefits, as individual contributions collectively ensure the health and sustainability of coastal ecosystems for all.

The framework also highlights the influence of SC, CV, and SE on CB, with direct and indirect pathways shown via solid and dashed arrows. The feedback loops from CB to SC and from CB to CV underscore the reinforcing effects of CB, where positive outcomes enhance SC and CV, fostering continued collaboration. These cyclical processes create a self-sustaining mechanism that supports ongoing cooperation of shared resource management. Ultimately, the research aims to analyze the interactions between SC, CV, SE, and CB, using insights from the PGE to inform strategies for fostering sustainable coastal resource management.

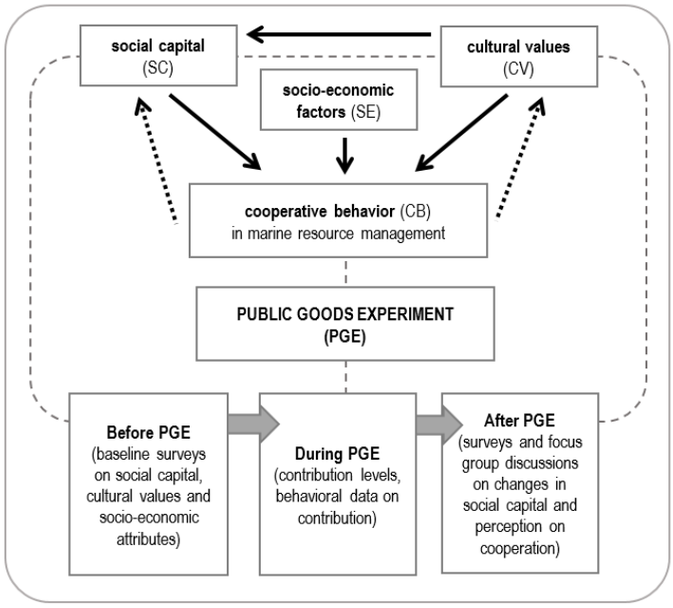


Figure 1: **Conceptual Framework**

Participant Selection. To ensure a representative and unbiased sample for the PGE, participants will be selected using a stratified random sampling method. The target population includes the indigenous and local (non-indigenous) fishers in the two study sites. The population in each study site will be divided into two strata based on group (indigenous or local), and participants will be randomly selected from each stratum to achieve balanced representation. Recruitment efforts will involve close collaboration with community leaders and organizations to foster trust and encourage participation. Strict ethical standards will be adhered to, ensuring informed consent is obtained from all participants, with a clear explanation of the study's purpose and their rights. This approach aims to generate comprehensive and reliable data, enabling us to analyze cooperative behavior, social capital and cultural values within the context of sustainable coastal resource management.

Group Composition and Assignment. The samples at each site will be divided into three groups: homogenous indigenous fishers' group (Group 1), homogenous local fishers' group (Group 2), and mixed indigenous and local fishers' group (Group 3). Participants from each stratum (indigenous or local) at each site will be randomly assigned to one of these three groups to ensure balanced group sizes. The total number of participants in each stratum will be divided by three, and participants will be allocated to groups to maintain equal group sizes. This ensures fair and unbiased comparison of cooperative behavior and social capital across different groups. The purpose of this group assignment is to compare the behavior of homogeneous and heterogeneous groups, focusing on social capital and cooperative behavior.

Experimental Design. The PGE will be conducted separately for each of the three groups at both research sites to prevent cross-group influence and ensure that the unique dynamics of each group are captured accurately. This will assume the outcomes and benefits of coastal resource conservation efforts as a public good, represented by the communal fund in the PGE. The experimental sessions will be the same for all groups, and will be done simultaneously but in different venues.

Each participant will receive an initial endowment (cash or token representing the resource units) per round. Participants will decide privately how much of this endowment they wish to voluntarily contribute to a communal fund, representing the outcomes and benefits of coastal resource conservation efforts in the shared marine resources. Contributions will be anonymous to encourage independent decision-making and to prevent social pressure from influencing contributions.

The total amount contributed to the communal fund by all participants in the group will be multiplied by a marginal per capita return. This multiplier reflects the shared benefit that each participant receives from the communal resource, regardless of their individual contribution. The resulting amount will then be divided equally among all participants, representing their payoff (including their earnings) for that round. At the end of each round, participants will receive their individual payoff, which includes their retained endowment and their share of the public good. The amount of the individual payoff will not be disclosed to other participants, ensuring privacy and preventing strategic behavior based on others' earnings.

The experiment session for each group will consist of several rounds or trials, allowing participants to observe and adjust their contribution decision strategies over time. After each round, participants will receive feedback on total contributions to the communal fund and the individual payoffs but will not be informed on specific contributions of others to maintain privacy and prevent direct influence. This repeated

interaction and feedback are crucial for understanding how social capital and cooperation develop or erode throughout the experiment, especially in response to others' cooperative behavior or contributions to the communal fund.

Data Collection Process. The data collection process will be carried out in three main phases: pre-experiment, during the public good experiment, and post-experiment.

1. **Pre-experiment Phase.** Before the public goods experiments, participants will complete a comprehensive survey designed to gather baseline data on socio-economic characteristics, social capital, cultural values, and expectations of cooperation. This phase will ensure that relevant demographic and socio-cultural information is collected, enabling comparisons and deeper analysis in later stages. *Data-gathering tools*: socio-demographic survey, social capital pre-experiment survey, cultural values pre-experiment survey.

2. **During the Public Goods Experiment.** Data will be collected by observing participant behavior and recording their decisions regarding contributions to the communal fund. Participants will be asked to document the amount they contribute, their rationale for contributing (or not contributing), and their perceptions of fairness within the group. This phase will capture actual data on cooperative behavior in a controlled experimental setting. The experiments will be conducted using standardized procedures to ensure consistency across groups and sites. *Data gathering tools*: endowment and contribution records, group observation notes, participant's recording forms.

3. **Post-experiment Phase.** After the public goods experiment, participants will complete a follow up survey to assess changes in their social capital, cooperative behavior, and cultural values. Additionally, focus group discussions (FGDs) will be conducted with selected participants to gain qualitative insights into their decision-making processes during the PGE, as well as their perceptions of the effectiveness and the role of cultural values in shaping cooperation. *Data-gathering tools*: social capital post-experiment survey, cultural values post experiment survey, interview notes.

As mentioned above, there are several data-gathering tools that will be designed and crafted. These include the survey questionnaires for socio-demographic characteristics, social capital, and cultural values of the participants in fostering sustainable coastal resource management.

- The socio-economic survey will gather essential background information about the respondents, including their age, gender, education level, household size, income, fishing experience, and reliance on marine resources for livelihood. This section will also assess perceptions of resource availability, competition, and population dynamics to provide context for understanding cooperative behavior. This will include participants' observations of changes in fish stocks, access to fishing areas, and competition for resources.
- The social capital survey will focus on measuring trust and reciprocity among fishers and their communities, with specific questions on trust in peers, regulatory authorities, and the enforcement of coastal resource regulations. It will also explore respondents' participation in group activities, decision-making processes, and perceptions of fairness and community support in sustainable coastal resource governance. It will also include questions that will elicit participants' support and compliance to existing sustainable fishing regulations, perceptions of enforcement effectiveness, and how these influence their cooperation.
- The cultural values survey will delve into the shared beliefs, traditions, and attitudes that shape respondents' approaches to coastal resource management, such as communal responsibility, respect for nature, and adherence to traditional ecological practices. This section will also capture insights into how cultural norms influence cooperation and conservation behavior of both indigenous and local perspectives.

FGDs are a critical qualitative tool in the study, providing in-depth insights into perceptions, behaviors, and attitudes that cannot be fully captured by surveys or experiments. Below are the key matters to be raised and information to be gathered:

1. Social Capital (Trust, Reciprocity and Social Network): Investigate trust levels among community members, reciprocity practices, and the role of social networks in cooperation and conflict resolution. Information to gather includes examples of trust and reciprocity in action, challenges in maintaining trust, and suggestions for strengthening social networks to improve cooperation.
2. Cultural Values and Social Norms: Explore the influence of traditional beliefs, practices, and cultural attitudes on behavior and cooperation in resource management. Information to gather includes examples of cultural practices supporting or conflicting with conservation strategies, influence of norms on behavior, and recommendations for integrating cultural practices into conservation efforts.
3. Feedback on Public Goods Experiment (PGE): Assess the accuracy, fairness, and lessons learned from the PGE in reflecting real-world resource management dynamics. Information to gather includes motivations for cooperative or competitive behavior, suggestions for improving the experimental design, and insights into its impact on understanding cooperation.

4. Community Dynamics in Mixed Groups: Understand perceptions, collaboration, and fairness between indigenous and non-indigenous fishers. Information to gather includes examples of conflicts or synergies and strategies to manage diverse group dynamics.
5. Trust in Regulators and Enforcement Systems: Examine perceptions of fairness, transparency, and effectiveness of resource management authorities. Information to gather includes levels of trust, examples of enforcement outcomes, fairness of rules, and recommendations for improving trust and enforcement.
6. Perceptions of Marine Protected Areas (MPAs): Evaluate views on the role, effectiveness, and challenges of MPAs in conserving biodiversity. Information to gather includes awareness of MPA benefits, challenges due to regulations, and their impact on livelihoods.
7. Awareness and Participation in Conservation Activities: Explore knowledge and involvement in conservation programs. Information to gather includes specific programs participants are aware of, barriers to engagement, and suggestions for increasing awareness and participation.
8. Effectiveness of Current Conservation Strategies: Assess perceptions of existing conservation regulations and community-led initiatives. Information to gather includes feedback on strategy effectiveness and suggestions for improvement.
9. Barriers to Conservation and Cooperation: Identify challenges to conserving resources and fostering cooperation. Information to gather includes specific barriers, examples of successful or failed initiatives, and strategies to overcome challenges.
10. Gender Roles in Conservation: Examine roles of men and women in resource management and barriers to equal participation. Information to gather includes examples of gender-specific roles, barriers, and solutions for equal participation.
11. Resource Scarcity and Population Size: Investigate perceptions of resource availability, competition, and population effects on management. Information to gather includes examples of scarcity-driven cooperation or conflict, adaptation strategies, and opinions on solutions.
12. Effect of Socio-Economic Factors on Cooperation: Analyze the effects of age, education, income, and fishing experience on cooperation. Information to gather includes socio-economic disparities as barriers to action and support needed to address them.
13. Governance and Institutional Support: Understand perceptions of governance, co-management, and institutional roles. Information to gather includes effectiveness of governance systems, inclusivity in decision-making, and recommendations for improvements.
14. Feedback on Incentive Mechanisms: Explore participants' views on financial, social, or environmental incentives to encourage cooperation. Information to gather includes suggestions for incentives and concerns about fairness or feasibility.
15. Policy and Actionable Recommendations: Gather feedback on policies and ideas for new or adjusted policies for sustainable resource management. Information to gather includes policy suggestions, alignment with local needs, and potential incentives to promote compliance.
16. General Feedback and Suggestions: Provide an open platform for participants to raise additional issues or concerns. Information to gather includes unanticipated challenges or opportunities in resource management.

To supplement the data gathered from social experiments, surveys, and focus group discussions (FGDs), the project will collect existing secondary data from key sources, including the Bureau of Fisheries and Aquatic Resources (BFAR), local government units (LGUs), and relevant stakeholders. Contact will also be established with the Marine Protected Area Support Network (MSN) to access their resources, such as the Philippine MPA Database. This database contains essential information on MPA size, location, management effectiveness, and governance structures. Additionally, reports from MSN on community participation, ecological health, resource availability, biodiversity status, and conservation activities will be reviewed for further insights. These secondary data sources, combined with FGDs with local stakeholders, will validate and enhance the findings by providing a broader context and deeper understanding of how governance structures influence community-level cooperative behaviors. The integration of these data sources will also enable the project to assess whether cooperative behaviors observed in the study align with co-management principles, such as shared responsibilities and decision-making between governing bodies and communities.

Data Analysis. This will employ a combination of econometric techniques and PGE-specific analyses to examine the relationships between social capital, cultural values, socio-economic factors, and cooperative behavior in the context of sustainable coastal resource management. The econometric analyses aim to rigorously examine the relationships between key variables such as social capital, cultural values, socio-economic factors, and cooperative behavior, providing a structured approach to understanding the underlying dynamics in coastal resource management. These will include data regression, mixed-effects models, and latent class analysis. On the other hand, the PGE-specific analyses will be used to explore the unique behavioral patterns and group dynamics that emerge during the PGE, offering deeper insights into cooperation and free-riding. These will include the extent of free-rider analysis and cluster analysis. The data analysis will be conducted using Stata and R for econometric modeling and statistical analysis. Moreover, a simple thematic analysis will be employed to process and interpret the data gathered from

focus group discussions. This approach ensures a comprehensive understanding of community perspectives, providing actionable recommendations for enhancing resource management.

Dissemination and Utilization of Results. To ensure that the study meaningfully contributes to sustainable coastal resource management and empowers target beneficiaries, the findings will be disseminated and utilized through the following mechanisms to ensure practical application and community engagement:

a) Policy Recommendations: Evidence-based policy briefs will be shared with LGUs, BFAR, and local stakeholders to enhance MPA management and promote cooperative behavior.

b) Community Workshops: Interactive workshops will present findings to local and indigenous fishers, which include translating insights into actionable strategies and gathering community feedback.

c) Training Sessions: Capacity-building sessions for fishers and MPA managers will focus on best practices for cooperative coastal resource management to enhance social capital, and integrate traditional knowledge into management frameworks.

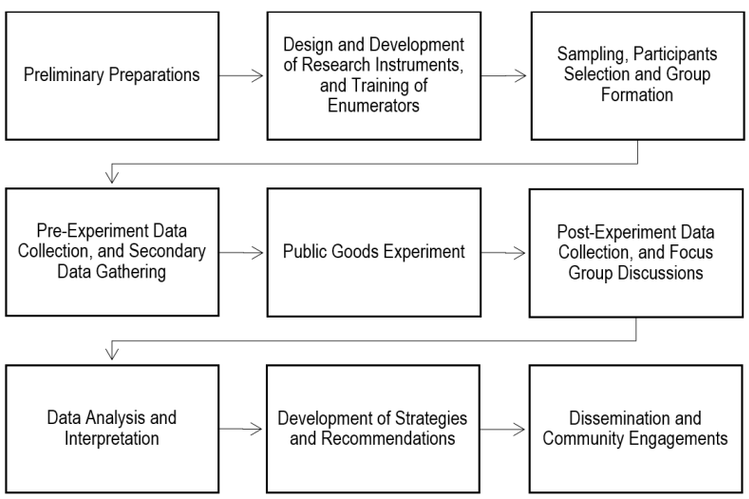
d) Action Plans: Using a participatory approach, action plans will be co-developed with the communities and stakeholders to institutionalize cooperative coastal resource management practices aligned with local policies and MPA frameworks.

e) Academic and Public Engagement: Findings will be shared through journal publications, conference presentations, and public awareness campaigns to inform academic, policy-making and general audiences.

f) Benchmarking Visits and Expert Consultations: Local and foreign benchmarking visits and consultations with experts will be conducted to integrate best practices, innovative strategies, and global insights into the research outcomes to ensure that findings are relevant and adaptable to varying contexts.

g) Production of a Manual: A comprehensive manual will be developed for conducting PGE in fishing communities to effectively manage common-pool fishery resources, providing practical guidelines and tools for researchers, practitioners, and policymakers.

Study Flow and Implementation: Figure 2 displays the flowchart outlining the sequential steps of this study, which centers on public goods experiments in coastal resource management. The process begins with preliminary preparations, which include defining objectives, outlining methodologies, logistic planning and coordinating with stakeholders. This is followed by the design and development of data collection instruments to ensure reliability and validity in data gathering. Participants are then selected, and groups are formed to engage in the economics experiments, where decision-making behaviors are observed. Pre-experiment and post-experiment activities include data collection and analysis, complemented by focus group discussions to gain deeper insights into participant behaviors and perspectives. The findings inform the development of strategies and recommendations to address the study's objectives, which are then shared through dissemination and community engagements to ensure practical applications. An optional reflective analysis and follow-up may also be conducted to evaluate the long-term impact and guide future research, ensuring a systematic and comprehensive approach to understanding public goods dynamics in coastal resource management.



Benchmarking Vi ıe Philippines face significant sustainability challenges, including limited community engagement, the free-rider problem, and insufficient integration of cultural values. Benchmarking coastal resource management practices in Japan, particularly in Kochi, offers valuable insights into addressing these issues. Japan's successful community-based resource management strategies, such as fishery cooperatives and the "satoumi approach," which harmonizes human activities with marine conservation, highlight the critical role of social capital in fostering community participation. By integrating traditional ecological knowledge, robust policy frameworks, and an

emphasis on trust and cooperation, Japan provides a model for enhancing the effectiveness of MPA management in the Philippines.

The visit aims to: (1) gain insights into community-based coastal resource management practices in Japan; (2) understand how social capital fosters cooperation in Japanese fishing communities; (3) identify strategies that integrate cultural values into marine conservation.; (4) Strengthen collaborative networks with Japanese institutions and coastal communities. The key activities to be conducted during the benchmarking include: (1) engaging with local fishing communities through community interviews to understand their cooperative behaviors, trust-building practices, and conservation methods; (2) collaborative meetings with Japanese researchers and practitioners to exchange knowledge on coastal resource management; (3) site visits to observe and document Japan's community-led coastal conservation activities; and (4) data analysis to compare practices and outcomes between Japanese and Philippine approaches to conservation.

The benchmarking visit will span six (6) days, including four (4) days allocated for travel to and from the destination. On Day 1, participants will arrive at Ninoy Aquino International Airport (NAIA) in Manila. Day 2 involves departure from NAIA and arrival in Kochi, Japan. Days 3 and 4 will focus on visits to coastal communities and research institutions, including the Tatsukushi Visitor Center for Ashizuri Uwakai Nature Park and Kashiwajima Island in Kochi, Japan. Activities during these visits will include collaborative discussions and interviews with fishers, community leaders, and officers-in-charge to explore cooperative behaviors and social capital. Day 5 will be dedicated to documentation, feedback, and team synthesis of findings, followed by travel from Kochi to Tokyo and onward to NAIA. On Day 6, the team will return to their official stations (Partido State University or Department of Agriculture-Bureau of Fisheries and Aquatic Resources Region 2).

To ensure the smooth execution and success of the benchmarking activity, specific roles and tasks have been assigned to the members. Each coordinator is responsible for overseeing critical aspects of the event, from travel logistics to research presentations and financial management. The roles of each team member is enumerated in Table 1.

Table 1. Roles and responsibilities of the team members during the benchmarking visit.

Team Member	Role	Responsibilities
Joela Mizchelle D. Dacillo (Project Leader)	Overall Event Coordination	Lead the delegation, coordinate with Japanese counterparts, facilitate meetings, and prepare trip reports.
Raul G. Bradecina (Project Staff)	Financial Management	Handle budget allocation and management, including processing expenses and reimbursements and ensuring financial documentation.
Emma L. Ballad (Project Staff)	Logistics and Field Coordination	Record sessions, take detailed notes, organize documentation, and assist with report preparation.
Garry P. Dacillo (Project Staff)	Travel, Meeting Arrangements, and Documentation	Manage travel logistics, arrange interviews and community visits, support on-site activities, and assist with report preparation.

The benchmarking visit is expected to generate several key outputs: (1) a comprehensive benchmarking report will be produced, providing detailed documentation of Japanese community-based conservation practices and their applicability to MPAs in the Philippines; (2) collaborative partnerships will be established and strengthened with Japanese institutions and communities; and (3) policy recommendations will be produced, offering evidence-based strategies for enhancing social capital and cooperative behavior, with insights that can be compared to the context of Philippine coastal communities.

(10) TECHNOLOGY ROADMAP (if applicable) (use the attached sheet)

(11) EXPECTED OUTPUTS (6Ps)

1. **Publication:** Prepare two (2) high-quality, publishable research articles, targeted for submission to high-impact peer-reviewed journals.
2. **Products/ Processes:** Develop a comprehensive manual for conducting public goods experiment (PGE) in fishing communities for managing common-pool fishery resources.
3. **People Services:** Provide training/workshops to at least ninety (90) extension beneficiaries focused on fostering social capital, promoting cultural values, and developing cooperative behavior to enhance sustainable coastal resource management practices.

4. **Places/ Partnerships:** Prepare and finalize six (6) memoranda of understanding (MOUs) and/or memoranda of agreement (MOAs) with local government units (LGUs) and barangays to ensure strong institutional partnership, active engagement, and collaboration.
5. **Policies**: Develop four (4) evidence-based policy recommendations, one tailored for each municipality involved in this study, focusing on sustainable management of shared fishery resources. These recommendations, derived from the findings of the study, will be submitted to target LGUs/ barangays for informed decision-making and enhanced coastal resource governance.

(12) POTENTIAL OUTCOMES

This study aims to generate actionable insights and tangible benefits for both local communities and policymakers, fostering sustainable marine resource management practices.

- Local and indigenous fishers will have enhanced capacities for managing MPAs through increased cooperation and the integration of social capital and cultural values into coastal resource management practices.
- Evidence-based recommendations from the study will lead to the adoption of new policies or the refinement of existing ones, focusing on community-based resource management and the preservation of marine biodiversity.
- Successful management strategies and cooperative frameworks developed during the project can be replicated in other coastal regions, contributing to broader improvements in MPA management across the country.

(13) POTENTIAL IMPACTS (2Is)

Social Impact: The study will elucidate the role of social capital and cultural values in fostering cooperative behavior within MPAs. By focusing on indigenous and local fishers, the research will enhance community engagement in coastal resource management, reinforcing trust and reciprocity essential for effective collaboration. This increased participation is expected to strengthen social cohesion and collective action, leading to more resilient and self-sustaining communities. Furthermore, integrating traditional ecological knowledge into conservation strategies will preserve cultural heritage while aligning with modern management practices, thereby reinforcing community identity and resilience.

Economic Impact: This study has the potential to improve the sustainability and productivity of marine resources, which are crucial to the livelihoods of coastal communities. By fostering effective MPA management through enhanced cooperation, the study aims to stabilize fish stocks and ensure long-term resource availability, directly benefiting local and indigenous fishers. Additionally, improved ecosystem health could boost ecotourism and related economic activities, diversifying income sources for these communities. The insights gained will inform policy decisions, leading to more efficient resource management and enhanced economic stability in coastal regions.

(14) TARGET BENEFICIARIES

This study will benefit a range of stakeholders by addressing key challenges in marine resource management and providing actionable insights for sustainable practices:
1. Indigenous and Local Fishers: The primary beneficiaries are the indigenous and local fishers in coastal communities of the Bicol Region and Cagayan Valley. By enhancing understanding of social capital and cooperative behavior, the study aims to empower these fishers to engage more effectively in MPA management, leading to sustainable resource use and improved economic stability.
2. Community Leaders and Organizations: Local leaders and organizations involved in MPA management will benefit from actionable insights, enabling them to implement strategies that align with community dynamics and cultural values, thereby enhancing management effectiveness.
3. Policy-makers and Conservation Practitioners*:* The study will provide evidence-based recommendations for policy-makers and conservation practitioners at regional and national levels, supporting the development of policies that foster sustainable coastal resource management and community resilience.
4. Academic and Research Institutions: The research will contribute to academic literature on social capital and marine resource management, offering valuable data and insights for future studies and conservation initiatives.

(15) SUSTAINABILITY PLAN

For the sustainability plan on this research project, we will prioritize on creating a long-term effect by actively involving local stakeholders throughout the study. We will collaborate with the local government units, community leaders, and non-governmental organizations to make sure that the findings and strategies which we will develop will be integrated into the existing coastal management programs. Capacity-building workshops will also be conducted to empower members with knowledge and tools for

sustained conservation efforts. The research will also involve a dissemination phase where results are shared with policy-makers and other similar coastal communities, thus providing a framework for replication and adaptation in similar contexts. Long-term impacts of the interventions will be followed up through continual monitoring, while strategies will be adapted to maintain and develop social capital for conservation.

(16) GENDER AND DEVELOPMENT (GAD) SCORE (refer to the attached GAD checklist)

16.66

(17) LIMITATIONS OF THE PROJECT

While this project is designed to provide valuable insights into the dynamics of social capital and cooperative behavior in MPAs, there are certain limitations to consider.

- The study focuses on specific coastal regions which may limit the applicability of the findings to other areas with different social and environmental contexts. Although the use of PGE offers a controlled method for analyzing cooperative behavior, it may not fully capture the complexities of decision-making in real-world scenarios.
- The timeline of the project may limit the ability to observe long-term behavioral changes and the sustainability of the interventions. Behavioral shifts in community cooperation and resource management may take longer to manifest than the project duration allows.
- The success of the project heavily relies on the active participation and cooperation of local and indigenous fishers. Any decrease in engagement due to external factors, such as economic pressures or environmental changes, could hinder data collection and the effectiveness of the interventions.
- The success of the project is contingent on the availability of financial and human resources throughout its duration. Any disruptions in funding or staffing could impede the project's progress and the implementation of planned activities.

Despite these limitations, the project is carefully designed to mitigate risks, with strategies in place to adapt methodologies as needed and ensure that the insights gained are robust and actionable. The expected outcomes, particularly in enhancing sustainable marine resource management, offer significant value that justifies the investment in this research.

(18) LIST OF RISKS AND ASSUMPTIONS RISK MANAGEMENT PLAN (List possible risks and assumptions in attaining target outputs or objectives.)

The possible risks are as follows:

1. There is a risk that local and indigenous fishers may not fully engage in the study due to external factors such as economic pressures or environmental changes. A lack of participation could negatively affect the quality and completeness of the data, thereby compromising the study's outcomes.
2. Focusing solely on the Bicol Region and Cagayan Valley may limit the generalizability of the findings. If the results are not applicable to other regions with different socio-cultural contexts, the broader impact of the project could be reduced.
3. The PGE used in the study may not fully capture the complexities of real-world decision-making in MPA management. This could result in findings that are not entirely reflective of actual community behavior, limiting the practical applicability of the results.
4. Unanticipated changes in environmental conditions or economic circumstances during the project period could disrupt the study's progress, potentially leading to incomplete data collection or altered participant behavior, which would negatively impact the study's outcomes.

The assumptions of this study are as follows:

1. Local and indigenous fishers will actively participate in the study, providing honest and thoughtful responses in surveys and experiments. This engagement is crucial for collecting accurate data and achieving meaningful insights.
2. The social capital and cultural values that will be examined in the targeted regions are representative enough to provide insights that can inform broader MPA management strategies.
3. It is assumed that PGE will effectively simulate and capture cooperative behavior in a way that reflects real-world dynamics within the communities.
4. The project assumes consistent support from community leaders and organizations throughout the study. This support is essential for successful participant recruitment, data collection, and the overall implementation of the research activities.
5. The study assumes that environmental and economic conditions will remain relatively stable during the research period, allowing for consistent community participation and reliable data collection.

(19) LITERATURE CITED

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(20) PERSONNEL REQUIREMENT		
Position	**Percent Time Devoted to the Project**	**Responsibilities**
Project Technical Assistant I	**100%**	Technical support for research activities, data collection and management, data analysis assistance, fieldwork support, technical documentation, training and capacity building support, compliance and ethical standards
Project Leader	**30%**	Overall project management, research design and oversight, stakeholder engagement, data analysis and reporting, compliance and ethics
Project Staff 3	**20%**	Research project implementation support, technical assistance, data management and analysis assistance, compliance and ethical standard, and other related tasks
Project Staff 2	**20%**	Research project implementation support, logistic coordination, data management and analysis assistance, compliance and documentation and other related tasks
Project Staff 1	**20%**	Research project implementation support, on-site study areas support, data collection, data management and analysis, documentation, and other related tasks

(21) BUDGET BY IMPLEMENTING AGENCY				
IMPLEMENTING AGENCY	**PS**	**MOOE**	**EO**	**Total**
Partido State University				
Year 1	1,087,624.00	1,520,617.00	120,000.00	**2,728,241.00**
Year 2	1,125,936.00	1,145,823.00	0.00	**2,271,759.00**
TOTAL	2,213,560.00	**2,666,440.00**	**120,000.00**	**5,000,000.00**

(22) OTHER ONGOING PROJECTS BEING HANDLED BY THE PROJECT LEADER: 0 (number)		
Title of the Project	**Funding Agency**	**Involvement in the Project**
N/A	**N/A**	**N/A**

(23) OTHER SUPPORTING DOCUMENTS (Please refer to page 2 for the additional necessary documents.)

I hereby certify the truth of the foregoing and have no pending financial and/or technical obligations from the DOST and its attached Agencies. I further certify that the programs/projects being handled are within the prescribed number as stipulated in the DOST-GIA Guidelines. Any willful omission/false statement shall be a basis of disapproval and cancellation of the project.

	SUBMITTED BY (Project Leader)	**ENDORSED BY (Head of the Agency)**
Signature		
Printed Name	**JOELA MIZCHELLE D. DACILLO, PhD**	**ARNEL B. ZARCEDO, EDEL**
Designation/Title	**Instructor I**	**SUC President III**
Date		

Note: See guidelines/definitions at the back.

DOST Form 2 (for Basic/Applied Research)
DETAILED R & D PROJECT PROPOSAL

I. General Instruction: Submit through the DOST Project Management Information System (DPMIS), http://dpmis.dost.gov.ph, the detailed R&D proposal for the component project together with the detailed proposal of the whole Program, project workplan, line-item budget (LIB), 1-page curriculum vitae of the Project Leader, and Certificate of Incorporation or DTI Registration (if applicable) and other applicable supporting documents required under item II.23 below. Also, submit four (4) copies of the proposal together with its supporting documents. Use Arial font, 11 font size.

II. Operational Definition of Terms:
1. Title- the identification of the Program and the component projects.

Project- refers to the basic unit in the investigation of specific S&T problem/s with predetermined objective/s to be accomplished within a specific time frame.

Project Leader- refers to a project's principal researcher/implementer.

Project Duration- refers to the grant period or timeframe that covers the approved start and completion dates of the project, and the number of months the project will be implemented.

Implementing Agency- the primary organization involved in the execution of a program/project which can be a public or private entity

2. Cooperating Agency/ies- refers to the agency/ies that support/s the project by participating in its implementation as collaborator, co-grantor, committed adopter of resulting technology, or potential investor in technology development or through other similar means.

3. Site/s of Implementation- location/s where the project will be conducted. Indicate the barangay, municipality, district, province, region, and country.

4. Type of Research- indicates whether the project is basic or applied.
Basic research- is an experimental or theoretical work undertaken primarily to acquire new knowledge of the underlying foundations of phenomena and observable facts, without any particular or specific application or use in view.
Applied research- is an investigation undertaken in order to utilize data/information gathered from fundamental/basic researches or to acquire new knowledge directed primarily towards a specific practical aim or objective with direct benefit to society.

5. R&D Priority Area and Program- based on the Harmonized National R&D Agenda 2017-2022, indicates which R&D agenda the project can be categorized in: Agriculture, Aquaculture and Natural Resources; Health; Industry, Energy, and Emerging Technology; Disaster Risk Reduction and Climate Change Adaptation; and Basic Research. Indicate also the specific Commodity/Sector, whether crops, livestock, forestry, agricultural resources or socio-economics; fisheries or aquatic resources; biotechnical, pharmaceutical, or health services; biotechnology, information technology, material science, photonics or space technology; industry, energy, utilities or infrastructure.

Sustainable Development Goal (SDG) Addressed- indicates which among the 17 SDGs adopted by the United Nations Members States are addressed by the project

6. Executive Summary- briefly discusses what the whole proposal is about

7. Introduction- a formally written declaration of the project and its idea and context to explain the goals and objectives to be reached and other relevant information that explains the need for the project and aims to describe the amount of work planned for implementation; refers to a simple explanation or depiction of the project that can be used as communication material.

7.1. Rationale- brief analysis of the problems identified related to the project
Significance- refers to the alignment to national S&T priorities, strategic relevance to national development and sensitivity to Philippine political context, culture, tradition and gender and development.

7.2. Scientific Basis- other scientific findings, conclusions or assumptions used as justification for the research
Theoretical Framework- the structure that summarizes concepts and theories that serve as basis for the data analysis and interpretation of the research data.

7.3. Objectives- statements of the general and specific purposes to address the problem areas of the project.

8. Review of Literature- refers to the following: (a) related researches that have been conducted, state-of-the-art or current technologies from which the project will take off; (b) scientific/technical merit; (c) results of related research conducted by the same Project Leader, if any; (d) Prior Art Search, and; (e) other relevant materials.

9. Methodology- discusses the following: (a) variables or parameters to be measured and evaluated or analyzed; (b) treatments to be used and their layout; (c) experimental procedures and design; (d) statistical analysis; (e) evaluation method and observations to be made, strategies for implementation (Conceptual/Analytical framework).

10. Technology Roadmap (if applicable)**-** a visual document that communicates the plan for technology. It is a flexible planning technique to support strategic and long-range planning by matching short- and long-term goals to specific technology solutions.

11. Expected Outputs (6Ps)- deliverables of the project based on the 6Ps metrics (Publication, Patent/Intellectual Property, Product, People Service, Place and Partnership, and Policy).

Publication- published aspect of the research, or the whole of it, in a scientific journal or conference proceeding for peer review, or in a popular form.

Patent/Intellectual Property- proprietary invention or scientific process for potential future profit.

Product- invention with a potential for commercialization.

People Service- people or groups of people, who receive technical knowledge and training.

Place and Partnership- linkage forged because of the study.

Policy- science-based policy crafted and adopted by the government or academe as a result of the study.

12. Potential Outcomes- refer to the result that the proponent hopes to deliver three (3) years after the successful completion of the project.

13. Potential Impacts

Social Impact- refers to the effect or influence of the project to the reinforcement of social ties and building of local communities.

Economic Impact- refers to the effect or influence of the project to the commercialization of its products and services, improvement of the competitiveness of the private sector, and local, regional, and national economic development.

14. Target Beneficiaries- refers to groups/persons who will be positively affected by the conduct of the project.

15. Sustainability plan- refers to the continuity of the project or how it shall be operated amidst financial, social, and environmental risks.

16. Gender and Development (GAD) Score- refers to the result of accomplishing GAD checklists (for project monitoring and evaluation/project management and implementation) to highlight the contribution of the project in the achievement of the objectives of Republic Act 7192, "Women in Development and Nation Building Act," interpreted as gender-responsive, gender-sensitive, has promising GAD concepts, or GAD is invisible.

17. Limitations of the Project- refer to restrictions or constraints in the conduct of the project.

18. Risk- refers to an uncertain event or condition that its occurrence has a negative effect on the project.

Assumption- refers to an event or circumstance that its occurrence will lead to the success of the project.

19. Literature Cited- an alphabetical list of reference materials (books, journals and others) reviewed. Use standard system for citation.

20. Personnel Requirement- details on the position of personnel to be involved in the project, percent time devoted to the project, and responsibilities.

21. Budget By Implementing Agency- personnel services (PS), maintenance and other operating expenses (MOOE), and equipment outlay (EO) requirement of the project by implementing agency for Year 1 and for the whole duration of the project. Please refer to the DOST-GIA Guidelines for the details (Section IX.B of DOST Administrative Order (A.O.) 011, s. 2020).

a. **PS**- total requirement for wages, salaries, honoraria, additional hire and other personnel benefits.

b. **MOOE**- total requirement for supplies and materials, travel expenses, communication, and other services.

c. **EO**- total requirement for facilities and equipment needed by the Program.

22. Other Ongoing Projects Being Handled By the Project Leader- list of ongoing projects being handled by the Project Leader funded by the DOST-GIA Program and other sources, and the accompanying responsibilities relevant to the project.

23. Other supporting documents required- as stated in Section VII of DOST A.O. No. 011, Series of 2020 – Revised Guidelines for the Grants-in-Aid Program:

a. Detailed breakdown of the required fund assistance to indicate the counterpart of the proponent and other fund sources including letter/s of commitment from the implementing, collaborating and coordinating agency/entity/ies;[1]

b. A counterpart fund, in kind and/or in cash, shall be required from the implementing agency/entity as one of the application requirements. All projects must have a minimum of 15% counterpart contribution except for projects involving public good;[1]

c. Curriculum Vitae or Personal Data Sheet (PDS) of Project Leader and other co-researchers/implementers. The service record may be requested if needed;[1]

d. Clearance from the DOST or the Funding Agency (e.g., DOST Councils) on previously funded completed projects handled by the Project Leader;[1]

e. Approval from the institution's ethics review board for research involving human subjects or in the case of animal subjects, approval from the Bureau of Animal Industry (BAI) (for PCAARRD- and PCHRD-monitored projects);

f. Clearance from the DOST Biosafety Committee (DOST-BC) shall be required for research proposals involving the use of GMOs under contained use (i.e., experiments done in laboratories, screen house, green house). For projects other than contained use, they shall be referred to the appropriate agency. The DOST Sectoral Councils, after determination as to whether or not the proposal has biosafety implications, shall endorse the same to the DOST-BC in accordance with the prescribed format under Annex 3 of the Philippine Biosafety Guidelines for Contained Use of Genetically Modified Organisms (series of 2014) (if applicable); and

g. For the private non-profit/non-government/people's organizations and startups:

i. Up-to-date Securities and Exchange Commission (SEC) registration, or Department of Trade and Industry (DTI) registration, or Cooperative Development Authority (CDA) registration certificate, or other authenticated copy of latest Articles of Cooperation and other related legal documents;
ii. Co-signers Statement (if applicable);
iii. Copy of latest Income Tax Return;
iv. Mayor's permit where the business is located;
v. Audited Financial Statements for the past three (3) years preceding the date of project implementation or in case of those with operation of less than 3 years, for the years in operation and proof of previous implementation of similar projects (or in the case of startups, at least for one (1) year);
vi. Document showing that NGO/PO has equity to 20 percent of the total project cost, which shall be in the form of labor, land for the project site, facilities, equipment and the like, to be used in the project;
vii. Disclosure of other related business, if any;
viii. List and/or photographs of similar projects previously completed, if any, indicating the source of funds for implementation;
ix. Sworn affidavit of secretary of the NGO/PO that none of its incorporators, organizers, directors or officers is an agent of or related by consanguinity or affinity up to the fourth civil degree to the official of the agency authorized to process and/or approved the proposed MOA, and release of funds;

h. For CSOs, compliance to regulations as required by the General Appropriations Act (GAA) pertaining to fund transfers to Civil Society Organizations (CSOs); and

i. For foundations, DOST certification as accredited by the Science and Technology Foundation Unit

[1] *required of all proposals*

III. Criteria for Evaluation:

A. **Criteria for Evaluating Proposals**

Criterion	Definition

Relevance or Significance	Aligned to national S&T priorities, strategic relevance to national development and sensitivity to Philippine political context, culture, tradition and gender and development
Technical / Scientific Merit	Sound scientific basis to generate new knowledge or apply existing knowledge in an innovative manner
Budget Appropriateness	The proposed budget is commensurate to the proposed work plan and deliverables.
Competence of Proponent	Proponent's expertise is relevant to the proposal and with proven competence to implement, manage and complete R&D programs/projects within the approved duration and budget.

B. **Governing Council / Board and EXECOM's Evaluation Criteria**

Criteria	Indicators	Raw Score
A. Soundness of Proposal (20%)	R&D addresses relevant sectoral need (applicable to pressing concern)	5
	Solution provided is most effective (compared to other proposed solutions)	5
	Proposed budget is reasonable (project is not expensive vis-a-vis output)	5
	Work plan is doable in a given timeframe	5
B. Suitability of Output (30%)	R&D output is cost-effective (cost is competitive in relation to new or existing products or process)	5
	Has identified partners to adopt the technology (with letter of support from the head of the company)	5
	Output can be commercialized (through an existing manufacturer, spin-off or start-up company)	5
	R&D utilization is timely (output should not be overtaken by other solutions)	5
C. Significance of Outcome (30%)	Economic: increase in productivity, increase in income, new jobs generated, high return of investment (ROI)	5
	Social: working partnerships established, training opportunities provided, policies adopted, increased access to basic services (i.e., food, health, education); political, cultural, gender sensitivity and inclusivity	5
	Environment: enhanced environmental health standards, no adverse effect to the environment	5
	Sustainability: sustainability mechanisms established in terms of institutional, financial and human resources capability (submission of a new proposal to sustain a completed or ongoing proposal does not constitute sustainability of the project)	5
D. Competence of Proponent (20%)	Proponent's expertise aligned with the proposal	5
	Collaboration with relevant agencies and/or industry partners	5
	Thorough understanding of the proposal's deliverables	5
	DOST has good experience with the proponent	5

C. **Additional Criteria on Gender and Development (GAD)**